The empirical recurrence for OEIS A237393: recovering a conjecture that is not written down

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Abstract

OEIS A237393 records "Empirical recurrence of order 97", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 217$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 97, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 97 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237393 is "Number of (n+1)X(2+1) 0..3 arrays with the maximum plus the upper median minus the lower median of every 2X2 subblock equal.". It has offset 1 and begins

$$1314, 11988, 114034, 1153778, 11725568, 121195320, 1254144270, 13055041806, \ \ldots$$

The entry states:

> Empirical recurrence of order 97 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 10:09:19, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 217$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 217.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 434$ exact terms of the model returns order 97 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{97} c_i\, a(n-i),$$

c_1	40	c_{26}	6233452907302495442124	c_{51}	11999955246962841684730550434
c_2	−328	c_{27}	−21848936013539181342008	c_{52}	15096517626640555561653832898
c_3	−7431	c_{28}	−46523103965394964177864	c_{53}	−2607414762788825257186391751
c_4	139522	c_{29}	252978210449289742594506	c_{54}	−1722033846791877339694206168
c_5	125044	c_{30}	220342419073727480271844	c_{55}	45074117995292705233903932244
c_6	−18314865	c_{31}	−2300120612855918527319873	c_{56}	12165630137836397904668847267
c_7	82351007	c_{32}	−5877476412468261104698 5	c_{57}	−6295173321209261296913168008
c_8	1203998763	c_{33}	16793898465481673084879066	c_{58}	1057363211202948372443659030
c_9	−10689441512	c_{34}	−10697003567271438074852698	c_{59}	71318050341743206653630123033
c_{10}	−38731645343	c_{35}	−99213449968331429176718593	c_{60}	−1774788774702032474708485426
c_{11}	701739226699	c_{36}	122868987074191318521460871	c_{61}	−6533640273727455133740952994
c_{12}	−82333779532	c_{37}	472296462355009414618357366	c_{62}	29850951144464209710708578803
c_{13}	−29484917239371	c_{38}	−889765070968998476670226873	c_{63}	47929017463188521228203803785
c_{14}	70225530548082	c_{39}	−1774982561449129111939393713	c_{64}	−3202377263601970339822007528
c_{15}	846670112642333	c_{40}	4860358469926963912691308269	c_{65}	−2760694568119122975239731116
c_{16}	−3750829581468884	c_{41}	4965652570002955237012855100	c_{66}	25427607182796019267788600965
c_{17}	−16494233644279995	c_{42}	−21136314735257450996042222965	c_{67}	11996596890079033942676478503
c_{18}	120001405721049137	c_{43}	−8304548181001865821375948113	c_{68}	−1560511738872614776680431683
c_{19}	187915996694562625	c_{44}	74757949968687678240339208052	c_{69}	−355055685459912019022140613
c_{20}	−2731104815630560858	c_{45}	−5392365551363700308853924817	c_{70}	7510182404913911230183437927
c_{21}	116564533846371329	c_{46}	−216718498210023571352155475612	c_{71}	4324110230751077334174962435
c_{22}	4680532211739083701 3	c_{47}	99272432463534983289906417028	c_{72}	−284027776840727207849452827
c_{23}	−5683138738443642897 1	c_{48}	514423137007153574304017326552	c_{73}	1998237381399558762469142732
c_{24}	−6167986149186413456 66	c_{49}	−421643251699478479860661682367	c_{74}	8393215265555156924473732300
c_{25}	1404167185828778894846	c_{50}	−990207137852597716616443156153	c_{75}	−146158217526971164307986173

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 97, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $97 + 4$ terms removed returns order 97 as well, so $\deg q_{\min} = 97 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $97+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 97 that this sequence satisfies — and that family turns out to have exactly one member.

References

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